

XRPL Validator — Daily Engineering Report • 2026-08-09

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 6 commits (xrpl-validator 6)

Engineering changes

- **fix(xrpl-ledger): rank payment strands by UPPER BOUND, not by a realised trial** (0f55ba5 , xrpl-validator) `payment.rs` 's round loop trialled every strand, measured what each pass actually delivered, and picked the best spent-per-delivered. That is the model this repo already REFUTED for offer crossing in 03c2cb9 -- `ActiveStrands::activateNext` sorts by `qualityUpperBound` best-first, DROPS any strand whose bound misses `limitQuality`, and the loop takes the first candidate whose pass is not rejected, `continue`-ing past ...
- **feat(xrpl-node): FFI_VALCHECK — the oracle rule now works at VALUE level** (c7d4625 , xrpl-validator) `PROBE: CLEAN` has only ever meant libxrpl touched the same OBJECTS mainnet did. It never said libxrpl wrote the same NUMBERS, because the entire parity verdict is a mutation-KEY comparison. So when 9bc95c8 made value divergences visible on the native side, there was no way to attribute them: 175 value hits and no verdict on whether any of them were ours.
- **fix(xrpl-ledger): token escrow — Cancel and Finish never released the token** (6ef30c4 , xrpl-validator) #106179351: 413 txs, 221 matched. `EscrowCancel attempted=192 MATCH=0 DIVERGE-MUT=192` -- every `EscrowCancel` in the ledger, all the same shape, ours 3 mutations against mainnet's 8. `FFI PROBE: CLEAN`, so ours.
- **fix(xrpl-node): prefetch NFT pages for NFTTokenModify — it is tt 61, not 25..=29** (5aad67e , xrpl-validator) `is_nft_tx_type` was `(25..=29).contains(&tt)` — Mint, Burn, CreateOffer, CancelOffer, AcceptOffer. `NFTTokenModify` (XLS-46) is tt 61: it POSTDATES that contiguous range, so a transaction whose entire job is to locate a token and rewrite its URI never had the owner's `NFTTokenPages` fetched. libxrpl's `nft::findToken` then searched a page set that was never loaded and returned `tecNO_ENTRY` on a token that plainly exists.
- **feat(xrpl-ledger): DX_CLAMP — and it retires the ceil_in flip for good** (9e475fe , xrpl-validator) The 2026-08-08 attempt to flip the taker-clamped XRP floor to a ceil regressed #105284280 (78/78 -> 77/78) and took #105091578 from 0 value hits to 21. The note left behind said the next attempt had to establish, first, whether the twelve one-drop-short residuals reach that branch at all.
- **fix(xrpl-ledger): the bridged XRP mid-leg must CEIL to whole drops, not truncate** (9bb5ceb , xrpl-validator) Autobridged crossing routes value through XRP, and that mid-leg has to land on whole drops. `cross_bridged` truncated it — `me_rescale(xrp, 0, false)` — so one leg moved a drop more than the other could absorb and the pass came out skewed in both directions at once.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Stealth Meta-Address Fields on AccountRoot** (privacy) Add two optional SFields to AccountRoot — sfStealthScanPubKey and sfStealthSpendPubKey (compressed secp256k1 points) — settable via AccountSet after a
- **Native Subscription Pull-Payment Ledger Object** (payments) Add a new owner-reserved ledger object ltSUBSCRIPTION (keylet::subscription(payer, sequence)) holding sfPayee, sfMaxAmount, sfPeriod (ledgers between
- **Weighted Multi-Asset AMM Basket Pools** (amm-defi) Extend AMMCreate with an optional STArray sfAssets (2–8 entries, each an STObject of sfAsset + sfWeight in basis points summing to 10000) gated by a n

Appendix — commits

Time	Repo	Commit	Subject
13:22	xrpl-validator	0f55ba5	fix(xrpl-ledger): rank payment strands by UPPER BOUND, not by a realised trial
15:17	xrpl-validator	c7d4625	feat(xrpl-node): FFI_VALCHECK — the oracle rule now works at VALUE level
21:59	xrpl-validator	6ef30c4	fix(xrpl-ledger): token escrow — Cancel and Finish never released the token
22:47	xrpl-validator	5aad67e	fix(xrpl-node): prefetch NFT pages for NFTTokenModify — it is tt 61, not 25..=29
22:50	xrpl-validator	9e475fe	feat(xrpl-ledger): DX_CLAMP — and it retires the ceil_in flip for good
23:53	xrpl-validator	9bb5ceb	fix(xrpl-ledger): the bridged XRP mid-leg must CEIL to whole drops, not truncate